

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

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3x3 CONFUSION MATRIX

True labels (y): $[A, A, A, B, B, C, C, C, C]$

Predicted labels ($\hat{y}$): $[A, B, A, B, C, C, B, C, A]$

Step 1:- Build Confusion matrix

True/Pred	A	B	C
A	2	1	0
B	0	1	1
C	1	1	2

Compute per class Metrics:-

..Class A:-

$$TP (A \rightarrow A) = 2$$

$$FP = 0 + 1 = 1 (C \rightarrow A)$$

→ which were not truly/actually A but were predicted A

$$FN = 1 + 0 = 1 (A \rightarrow B)$$

→ which were actually A but were predicted B or C

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{2}{2 + 1} = 0.667$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{2}{2 + 1} = 0.667$$

$$F1 = \frac{2 \times 0.667 \times 0.667}{0.667 + 0.667} = 0.667$$

..Class B:-

$$TP = 1 (B \rightarrow B)$$

$$FP = 1 (A \rightarrow B) + 1 (C \rightarrow B) = 2$$

→ which were not actually B but were predicted B

$$FN = 0 + 1 (B \rightarrow C)$$

→ which were actually B but were predicted A or C

$$\text{Precision} = \frac{1}{1 + 2} = 0.333$$

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$$\text{Recall} = \frac{1}{(1+1)} = 0.5$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 0.4$$

• - Class C :-

$$TP = 2 (C \rightarrow C)$$

$$FP = 0 + 1 = 1 (B \rightarrow C)$$

↳ which weren't actually C but were classified/predicted as C.

$$FN = 1 (C \rightarrow A) + 1 (C \rightarrow B) = 2$$

↳ which were actually C but were predicted as A or B.

$$\text{Precision} = \frac{2}{(2+1)} = 0.667$$

$$\text{Recall} = \frac{2}{(2+2)} = 0.5$$

$$F1 = 0.571$$

Step 3 :- Average Metrics

$$\text{Macro-average Precision} = \frac{(0.667 + 0.333 + 0.667)}{3} = 0.556$$

$$\text{Macro-average Recall} = \frac{(0.667 + 0.5 + 0.5)}{3} = 0.556$$

$$\text{Macro-average F1} = \frac{(0.667 + 0.4 + 0.571)}{3} = 0.546$$

Step 4 :- Accuracy & Error Rate

$$\text{Total correct predictions} = 2 + 1 + 2 = 5$$

$$\text{Accuracy} = 5/9 \approx 55.6\%$$

$$\text{Error Rate} = 1 - 0.556 = 44.4\%$$

Example Dataset

True labels (y):

[A, A, A, B, B, C, C, C, C]

Predicted labels ($\hat{y}$):

[A, B, A, B, C, C, B, C, A]

Step 1. Build Confusion Matrix

True \ Pred	A	B	C
A	2	1	0
B	0	1	1
C	1	1	2

Step 2. Compute Per-class Metrics

♦ Class A

- $TP = 2$ ($A \rightarrow A$)
- $FP = 0 + 1 = 1$ ($C \rightarrow A$)
- $FN = 1$ ($A \rightarrow B$)
- $Precision = 2 / (2 + 1) = 0.667$
- $Recall = 2 / (2 + 1) = 0.667$
- $F1 = 0.667$

♦ Class B

- $TP = 1 (B \rightarrow B)$
- $FP = 1 (A \rightarrow B) + 1 (C \rightarrow B) = 2$
- $FN = 1 (B \rightarrow C)$
- $Precision = 1/(1 + 2) = 0.333$
- $Recall = 1/(1 + 1) = 0.5$
- $F1 = 0.4$

♦ Class C

- $TP = 2 (C \rightarrow C)$
- $FP = 0 + 1 = 1 (B \rightarrow C)$
- $FN = 1 (C \rightarrow A) + 1 (C \rightarrow B) = 2$
- $Precision = 2/(2 + 1) = 0.667$
- $Recall = 2/(2 + 2) = 0.5$
- $F1 = 0.571$

Step 3. Average Metrics

- **Macro-average Precision** = $(0.667 + 0.333 + 0.667)/3 = 0.556$
- **Macro-average Recall** = $(0.667 + 0.5 + 0.5)/3 = 0.556$
- **Macro-average F1** = $(0.667 + 0.4 + 0.571)/3 \approx 0.546$

Step 4. Accuracy & Error Rate

Total correct predictions = $2 + 1 + 2 = 5$

Total = 9

$$Accuracy = 5/9 \approx 55.6\%$$

$$Error Rate = 1 - 0.556 = 44.4\%$$

✓ This shows how **precision, recall, F1, accuracy, error rate** are computed directly from a dataset's predictions.
